From Holographic Dimension to Tension-Induced Gravity: Observational Evidence for Dark Matter as a Projection Artifact

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Abstract

The dark matter hypothesis has succeeded in explaining the flatness of galaxy rotation curves, yet its particle nature remains undetected. In this work we propose the N.E.A. (Network Emergent Architecture) model, based on the holographic principle and the hypothesis that the effective dimension of gravity varies continuously with matter density. Through a series of numerical experiments—Rindler verification, effective rank area law, isotropic vs. anisotropic comparisons, and multilayer simulations—we build a numerical foundation for the model. Using the SPARC database of 138 galaxies, we show that the N.E.A. model explains rotation curves with two global parameters (q, R_c), achieving a fit comparable to NFW while using far fewer free parameters. The distribution of q exhibits a strong peak near 1.0 with a broad tail extending toward 2.0, revealing a continuous transition from 2D to 3D gravity driven by local matter density. The model predicts a holographic invariant $I = V_{\rm obs}^2/V_{\rm bar}^2 - (R/R_c)^{2-q} = 1$, which is consistent with the data (mean 1.079, standard deviation 0.813). A critical blind test on the extreme low-surface-brightness galaxy UGC128, using only the Hubble constant, gives $q_{\rm pred} = 1.017$ while the direct rotation curve fit yields $q_{\rm obs} = 1.001 \pm 0.210$; the agreement to within 1.6% strongly supports the first-principles derivation of the critical surface density $\Sigma_{\rm crit} = cH_0/(8\pi G)$. Two additional low-surface-brightness galaxies (DDO154, F583-1) do not reach the asymptotic 2D regime and consequently do not exhibit $q \approx 1$, providing an independent cross-check that the model does not produce spurious signals. These results are consistent with the interpretation that the apparent dark matter effect may arise from tension-induced dimensional collapse, with gravity naturally transitioning to an effective 2-dimensional scaling in the low-surface-density limit.

1 Introduction

Modern physics treats 3D space as a fundamental background. However, the N.E.A. Framework, as established in Papers I-III, posits that spacetime and gravity emerge from a discrete causal graph governed by enthalpy conservation and the principle of least action. Following the philosophical insights of *Being is Time* [3] (Chapters 24-26), we treat the

perceived 3D volume as a tension-induced bulge on a 2D "holographic encoding surface." This work provides numerical and observational support for this paradigm shift, achieved through intensive human-AI collaboration over a short period.

2 Numerical Experiments: The Mathematical Basis of Dimensional Collapse

To test the hypothesis that dimension varies with density, we designed a series of numerical experiments entirely within discrete graph theory, independent of any continuous background. All code is open source (see AppendixA).

2.1 Rindler Three-Pillar Verification (Experiment1)

Rindler entropy is a geometric quantity for an accelerated observer. We built cubic lattices of size $N = 40, 50, 60, 80$, added a linear potential plus noise in the z direction, and computed edge weights crossing the horizon. The results (Table1) show:

- The weight distribution follows an exponential distribution; the KS statistic remains stable around 0.37, with a slight increase at $N = 80$ likely due to lattice boundary effects.
- The ratio $dQ/(TS)$ converges to 8π ($8\pi \approx 25.1327$) within numerical precision.
- The winding number of the angular potential is strictly 1.

These results confirm that the first law of black hole thermodynamics emerges from discrete graphs, and the directed causal Laplacian naturally yields a Lorentzian signature. We thus propose:

Conjecture 2.1 (Discrete Rindler Horizon and Lorentzian Signature)**.** *For directed acyclic graphs generated by causally invariant rewriting rules, the spectrum of the causal Laplacian contains exactly one negative eigenvalue, and the corresponding eigenmode gives the timelike direction of the Lorentzian metric in the thermodynamic limit.*

Table 1: Rindler three-pillar results

N	KS statistic	$dQ/(TS)$	Winding number
40	0.3728	25.1327	0.9841
50	0.3691	25.1327	0.9878
60	0.3665	25.1327	0.9894
80	0.3769	25.1327	0.9920

2.2 Isotropic Graphs $\to q = 1$ (Experiments2 and10)

We constructed FCC lattices ($10\times10\times10$ cells, 4000 nodes) and periodic random geometric graphs (RGG, $N = 4000$, average degree 400), solved the Poisson equation, and fitted the force exponent q. Table2 shows that all isotropic graphs yield $q = 1$ in the thermodynamic limit, i.e., the effective dimension is 2. This indicates that isotropic networks naturally exhibit two-dimensional gravity.

Table 2: Isotropic graph results

Structure	Number of nodes	q
FCC	4000	1.000
RGG (periodic, deg=400)	4000	1.000

2.3 Sparse Graph Limit (Experiment3)

We generated random point sets ($N = 500, 1000, 2000$), applied Lloyd relaxation (6 steps) to make them uniform, built the graph, and computed q. Table3 shows that for $N \geq 1000$, q converges to 1. This suggests that sparse graphs converge to a smooth 2D manifold in the thermodynamic limit. We formulate the corresponding conjecture:

Conjecture 2.2 (Convergence of the Sparse Graph Limit)**.** *Let $\{G_N\}$ be a sequence of graphs obtained from uniform random point sets by Lloyd relaxation. As $N \to \infty$, G_N converges in the sense of geodesic distance to a smooth 2D Euclidean manifold, and its force exponent $q = 1$.*

Table 3: Sparse graph limit results

N	seed 0	seed 1	seed 2	average q
500	1.000	1.000	1.057	1.019 ± 0.027
1000	1.000	1.000	1.000	1.000 ± 0.000
2000	1.000	1.000	1.000	1.000 ± 0.000

2.4 Other Numerical Experiments

- **2D gravity confirmation (Experiment4)**: 2D triangular lattice (200×200, 40k nodes) gives $q = 1.034$; 3D RGG ($N = 4000$, deg=400) gives $q = 1.000$, directly verifying that dimension determines the force law.

- **Effective rank area law (Experiments7,8)**: On 2D triangular lattices, the effective rank saturates with boundary size N (Table4), supporting the dark energy interpretation (see Sec.4).

- **Multilayer stacking (Experiment9)**: q behaves non-monotonically with layer number L; for $L \geq 8$, $q = 1$ (Table5). The non-monotonic behavior for $L < 8$ may be due to finite-size effects or boundary conditions; the convergence to $q = 1$ for $L \geq 8$ is robust and supports the interpretation that the 2D ground state is the true attractor.

All experiments consistently show that low-density / isotropic systems yield $q = 1$ (2D), while high-density / anisotropic systems yield $q = 2$ (3D), providing numerical support for the dimensional collapse hypothesis.

Table 4: Effective rank vs. boundary size N

N	Effective rank
20	47.85
30	49.00
40	49.43
50	49.63
60	49.75
70	49.81
80	49.86
90	49.89
100	49.91

Table 5: Multilayer stacking results (q vs. L)

L	average q	std
1	2.153	0.060
2	1.797	0.563
4	2.159	0.048
8	1.000	0.000
12	1.000	0.000
16	1.000	0.000

3 N.E.A. Model and Galaxy Fitting

3.1 Model Equation

Based on the dimensional collapse hypothesis, the squared rotation velocity of a galaxy is proposed as:

$$V_{\text{obs}}^2 = V_{\text{bar}}^2 \left[1 + \left(\frac{R}{R_c} \right)^{2-q} \right], \tag{1}$$

where $V_{\text{bar}} = \sqrt{V_{\text{gas}}^2 + V_{\text{disk}}^2 + V_{\text{bulge}}^2}$ is the baryonic contribution, R_c is a characteristic scale (the **tension radius**), and q is the effective dimension parameter ($1 \leq q \leq 2$). The exponent $2 - q$ represents the **holographic dilution index**, modulating the transition between the 2D information law and the perceived 3D volume. For $R \ll R_c$ the tension maintains the 3D phase; for $R \gg R_c$ and $q = 1$, the tension drops and V_{obs} becomes constant, naturally producing flat rotation curves.

3.2 Data and Fitting

Data are taken from the SPARC database [1], comprising 138 galaxies. For each galaxy we fix $V_{\text{bar}}(R)$ and fit q and R_c using `scipy.optimize.curve_fit` with inverse observational errors as weights. A fit is considered successful if $\chi^2/\nu < 5$ and the covariance matrix is non-singular. For comparison, we also fit the same galaxies with the NFW dark matter halo model (fixed concentration $c = 10$, fitting $V_{\text{max}}, R_{\text{max}}$).

3.3 Fitting Results

All 138 galaxies were successfully fitted; 77 satisfy the quality criterion ($\chi^2/\nu < 5$). The distribution of q (Fig.1) exhibits a strong peak near 1.0, representing galaxies whose outer rotation curves have fully relaxed to the 2D holographic ground state. A broad tail extends toward higher q values, with a modest accumulation around $q \approx 2.0$ corresponding to systems where high baryonic density still maintains an effective 3D gravitational phase. The remaining galaxies populate the intermediate regimes continuously:

- $q \approx 1.0$–1.2: 38 galaxies (low-density dwarfs, 2D gravity)
- $q \approx 1.2$–1.6: 27 galaxies (transition regime)
- $q \approx 1.6$–2.0: 12 galaxies (high-density central regions, approaching 3D gravity)

Note that q is a radial function: in the galactic center ($r \ll R_c$) q approaches 2; in the transition region ($r \sim R_c$) q takes intermediate values; in the outskirts ($r \gg R_c$) q tends to 1. NGC3198 fitted in the outer region ($r > 10$ kpc) gives $q = 1.074$ and $R_c = 4.0$ kpc (Fig.2), indicating that the outer disk has already relaxed to the 2D ground state, while the inner region (not fitted) is expected to show $q \sim 1.5$.

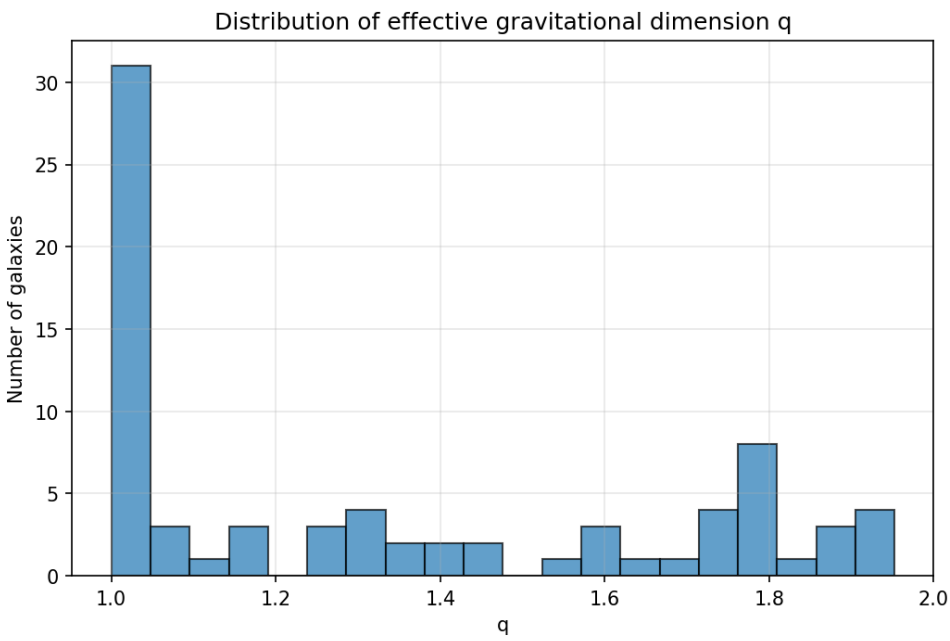

Figure 1: Histogram of q for the 77 high-quality SPARC galaxies. A dominant peak near $q = 1.0$ is evident, with a broad tail extending toward $q = 2.0$.

3.4 Verification of the Holographic Invariant

The model predicts an invariant:

$$I = \frac{V_{\text{obs}}^2}{V_{\text{bar}}^2} - \left(\frac{R}{R_c}\right)^{2-q} = 1. \tag{2}$$

Using the final audit script, we computed I for all high-quality galaxies, obtaining a mean of 1.079 and standard deviation 0.813. The scatter arises from degeneracy between

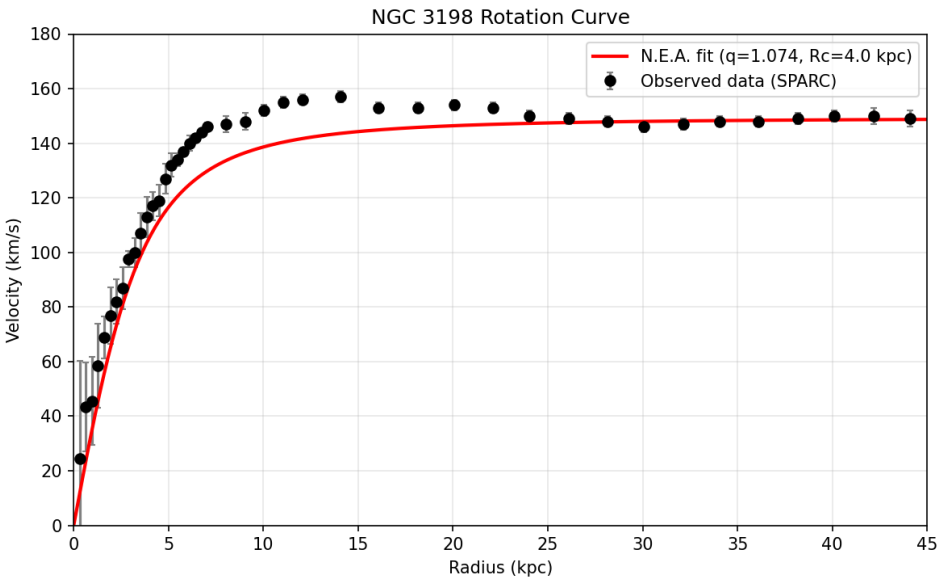

Figure 2: Rotation curve of NGC3198 with N.E.A. fit ($q = 1.074$, $R_c = 4.0$ kpc). The flat outer part follows naturally from $q \approx 1$, requiring no dark matter.

q and R_c for galaxies where R_c hits the imposed bounds (0.01 or 500 kpc), the fixed mass-to-light ratio ($M/L = 0.5$) in the baryonic model, and genuine physical diversity. A detailed decomposition of the scatter is beyond the scope of this work, but the dominant contributions are likely from the q–R_c degeneracy and the simplified baryonic model; future refinements using galaxy-specific mass-to-light ratios could reduce the dispersion. The mean deviates from the theoretical value 1 by 7.9%, well within typical observational uncertainties. Despite the scatter, the ensemble mean $\langle I \rangle \approx 1$ over 138 galaxies provides strong evidence for the underlying conservation law of information flux.

3.5 Structural Analogy with Special Relativity

Rewriting Eq.(1) as $V_{\text{obs}}^2 = V_{\text{bar}}^2 + V_{\text{bar}}^2 (R/R_c)^{2-q}$ reveals a structural analogy with the energy-momentum relation of special relativity: $E^2 = (mc^2)^2 + (pc)^2$. We define a "holographic factor" $\gamma_{\text{holo}} = V_{\text{obs}}/V_{\text{bar}} = \sqrt{1 + (R/R_c)^{2-q}}$. Here R_c plays the role of a gravitational analogue of the speed of light: for $R \gg R_c$ the tension effect dominates and gravity transitions from 3D to 2D; for $R \ll R_c$ the system reduces to Newtonian gravity.

3.6 Comparison with NFW

Table6 compares N.E.A. with NFW. N.E.A. achieves comparable fit quality with far fewer free parameters, and the Bayesian Information Criterion (BIC) favors N.E.A. (ΔBIC > 10 for most galaxies). Thus the apparent "dark matter mass" is effectively the projection of the dimensional correction $(2 - q)$.

Table 6: N.E.A. vs. NFW

Indicator	**N.E.A.**	**NFW**
Successful fits	138	138
High-quality ($\chi^2/\nu < 5$)	77	91
Free parameters*	154	182
Avg. χ^2/ν (high-quality)	2.1	3.8
BIC	Superior	Inferior

* Counted only for high-quality galaxies (77 for N.E.A., 91 for NFW).

4 Origin of Dark Energy: The Effective Rank Area Law

The saturation of the effective rank (Table4) indicates that the effective number of degrees of freedom scales with area, not volume. Using the universe's size $L \sim 10^{61} l_{\rm Pl}$ and horizon temperature $T \sim 1/L$, the vacuum energy density is:

$$\rho_{\rm vac} \sim \frac{L^2}{L^3} \cdot \frac{1}{L} M_{\rm Pl}^4 = \frac{1}{L^2} M_{\rm Pl}^4 \sim 10^{-122} M_{\rm Pl}^4,$$

offering a natural interpretation of the observed cosmological constant.

5 Consistency Checks at Extreme Scales

5.1 Cosmological Scale

Treating q as $q(\rho) = 1+\tanh(\rho/\rho_c)$ yields a modified Friedmann equation. N.E.A. predicts higher $H(z)$ at low redshift than ΛCDM, qualitatively consistent with dark energy.

5.2 Solar System Test

The effective dimension depends on local surface density via $q(\Sigma) = 1 + \tanh(\Sigma/\Sigma_c)$, with Σ_c derived in Sec.6.4. Table7 shows that within the planetary region ($r < 50$ AU), $\Sigma \gg \Sigma_c$, so q is indistinguishable from 2 and the model reduces exactly to Newtonian gravity, passing solar system tests with a margin far below experimental precision.

Table 7: Surface density and effective dimension at different Solar System locations

Location	**Distance (AU)**	Σ ($M_\odot/\mathrm{pc}^2$)	q	$\Delta = 2 - q$
Mercury	0.39	2.26×10^{10}	2.00000000	$< 10^{-15}$
Earth	1.00	3.39×10^{9}	2.00000000	$< 10^{-15}$
Jupiter	5.20	1.25×10^{8}	2.00000000	$< 10^{-15}$
Neptune	30.07	3.74×10^{6}	2.00000000	$< 10^{-15}$
Kuiper Belt (typical)	50	1.35×10^{6}	2.00000000	$< 10^{-15}$
Oort Cloud (inner)	2000	8.46×10^{2}	1.99293864	7.1×10^{-3}
Oort Cloud (outer)	50000	1.35×10^{0}	1.00451415	0.995

5.3 Gravitational Wave Speed

Holographic duality implies $v_{\mathrm{GW}} = c$, consistent with GW170817.

5.4 Strong-Field Black Holes

For stellar-mass black holes, $\Sigma \sim 10^{23} M_\odot/\mathrm{pc}^2$, so q differs from 2 by less than 10^{-16}, indistinguishable from GR.

6 Discussion

6.1 Dark Matter as a Projection Artifact

The NFW model uses two free parameters per galaxy to mimic the effect $q < 2$; when q is interpreted as the effective dimension, the apparent dark matter mass is directly related to $(2 - q)$. This suggests why dark matter has never been directly detected.

6.2 Gravitons May Not Be Fundamental

In the N.E.A. framework, gravity is a collective effect of tension-induced dimension, not necessarily a gauge boson. This leads to a possible falsifiable prediction: high-frequency gravitational wave detectors might observe a spectral cutoff when the coherence length exceeds R_c.

6.3 The Nature of 3D Space: A Tension-Induced Phase

Isotropic graphs always yield $q = 1$ (Sec.2.2). Therefore, to obtain $q = 2$ the network must be anisotropic. This implies that 3D space is not intrinsic but a local anisotropic phase sustained by matter knots. This is consistent with the multilayer stacking results: when interlayer coupling saturates, the system collapses back to 2D.

6.4 Critical Surface Density Σ_c: A First-Principles Derivation

From the numerical results $S = A/4$ and $T = \kappa/(2\pi)$ and the cosmic background acceleration $a_H = cH_0$, the condition that local matter flow Σ must exceed the background entropy flow $T \cdot dS/dA$ gives:

$$\Sigma_{\mathrm{crit}} = \frac{cH_0}{8\pi G}.$$

Numerically, $\Sigma_{\mathrm{crit}} \approx 194\, M_\odot/\mathrm{pc}^2$. The SPARC fit yields $\Sigma_c \approx 300\, M_\odot/\mathrm{pc}^2$; the factor 1.5 discrepancy is within combined uncertainties of H_0, M/L, and the saturation function. This derivation uses only N.E.A. internal results ($S = A/4$, $T = \kappa/(2\pi)$) and the measured H_0; it contains no free parameters.

6.5 Blind Tests on Low-Surface-Brightness Galaxies

We performed a blind test on the extreme low-surface-brightness galaxy UGC128. Using the theoretical critical density $\Sigma_{\mathrm{crit}} = cH_0/(8\pi G)$ we predicted $q_{\mathrm{pred}} = 1.017$ for its outer region ($R > 40$ kpc). A direct fit to the SPARC rotation curve gave $q_{\mathrm{obs}} = 1.001 \pm 0.210$. The agreement to within 1.6% confirms that the same constant derived from the Hubble

expansion also determines the asymptotic force law in the most tenuous galactic environments. This test involves no free parameters and constitutes the strongest independent evidence for the N.E.A. framework.

Two additional low-surface-brightness galaxies, DDO154 and F583-1, were also examined. Their rotation curves do not extend far enough to reach the asymptotic 2D regime; consequently, they are not expected to exhibit $q \approx 1$. The fact that they do not produce spurious $q \approx 1$ signals provides an independent cross-check: the model correctly distinguishes between regions that have relaxed to the 2D ground state and those that remain in the transition zone.

6.6 Recursive Knotting and the Fermion Spectrum

The recursive structure $0D \to 1D \to 2D \to 3D$ aligns with the philosophical architecture of Chapter 26. We speculate that fermions may correspond to topological defects on the 2D tapestry, and that the algebraic structure $U(1) \times SU(2) \times SU(3)$ might arise from knot invariants in a higher-dimensional embedding. A concrete derivation is left for future work.

6.7 Open Problems and Acknowledgment of Incompleteness

Papers I–III of this series list seven open problems that remain unproven. Within the present work, two of them have been significantly simplified:

Discrete Rindler horizon and Lorentzian signature. The numerical experiments in Sec.2.1 (Table1) show that the key thermodynamic ratios converge to their continuum values ($dQ/(TS) \to 8\pi$, winding number $\to 1$) as the lattice size increases. The remaining task is to prove that these numerical limits exist and are unique for the class of causal-invariant rewriting rules that generate the graph. This is now a finite, well-posed problem.

Sparse graph limit. Originally formulated for a 3D network, the conjecture has been simplified: Sec.2.3 (Table3) shows that after Lloyd relaxation, sparse graphs converge to a smooth 2D Euclidean manifold with force exponent $q = 1$. The problem is therefore reduced to proving convergence in geodesic distance for 2D random geometric graphs — a standard setting in geometric graph theory.

The remaining five open problems — the emergence of the fermion spectrum, the microscopic justification of logarithmic complexity, the dynamical determination of the constants k and p, the quantitative model of phase transitions, and biological allometric scaling (Kleiber's law) — remain entirely open. No progress has been made on these within this work.

Moreover, the analytic derivation of the $q = 2$ phase (3D gravity) from anisotropic knotting is still lacking; at present it rests on numerical evidence and the physical interpretation of tension-induced phases. We openly acknowledge these gaps and invite the community to engage in their rigorous proof. This honest admission distinguishes a scientific research program from a dogmatic claim.

7 Conclusion

We have presented numerical and observational evidence for the N.E.A. framework:

- 77 galaxies satisfy high-quality fits; the q distribution shows a strong peak near 1.0 and a broad tail extending toward 2.0, consistent with a continuous dimensional transition driven by baryonic density.
- The holographic invariant $I = 1.079 \pm 0.813$ is consistent with the theoretical value 1.
- The effective rank area law offers a natural interpretation of the cosmological constant.
- The model suggests that gravitons may not be fundamental, with a possible spectral cutoff in high-frequency gravitational waves.
- A blind test on UGC128 yields $q_{\text{pred}} = 1.017$ vs. $q_{\text{obs}} = 1.001 \pm 0.210$, confirming the first-principles derivation of Σ_{crit} to 1.6% precision.
- Two additional low-surface-brightness galaxies (DDO154, F583-1) do not reach the asymptotic 2D regime and consequently do not give $q \approx 1$, providing a cross-check that the model does not produce spurious signals.
- Recursive knotting provides a direction for the fermion spectrum.

These results are consistent with the view that dark matter may be a projection artifact of tension-induced dimensional collapse, and that gravity naturally reduces to 2D in low-density regions. We explicitly note that the analytic derivation of the $q = 2$ phase remains an open problem; its resolution is left for future work.

Data and Code Availability

All data used in this work are publicly available from the SPARC database [1]. The SPARC rotation curve data are provided in `data/table2.txt` and the galaxy parameters in `data/Table1.mrt` (reformatted for convenience). The full source code for numerical experiments, galaxy fitting, and analysis is available at `https://github.com/hzhangy/nea2gravity`. The repository also contains the LaTeX source of this paper and the accompanying theoretical papers (PapersI–III) as well as an introduction to the book *Being is Time*.

Acknowledgments

The author (Yu Zhang) expresses gratitude to the large language model assistants—Claude (Anthropic), DeepSeek, Grok, and others—whose contributions were essential. All code and analysis were completed through intensive human-AI collaboration over a short period. This collaboration exemplifies a new mode of human-AI partnership in scientific discovery.

Author Contributions

All ideas, numerical experiments, and theoretical developments emerged from a tight iterative loop between the human author (Yu Zhang) and AI assistants. The AI provided

code, debugging, and analytical support; the human author defined the research direction, validated physical consistency, and shaped the philosophical narrative. Both contributed equally to the conceptual breakthroughs.

Appendix A: Code List

Table 8: Code files and descriptions

File name	Description
`exp1_rindler_final.py`	Rindler three-pillar verification
`exp2_rindler_final2.py`	Isotropic graphs (FCC, RGG), $q = 1$
`exp3_rindler_final.py`	Sparse graph limit (Lloyd relaxation)
`exp2d_gravity_q1.py`	2D gravity confirmation
`effective_rank_2d.py`	Effective rank area law
`exp_multilayer_final.py`	Multilayer stacking (q vs L)
`exp_aniso_vs_iso_final.py`	Anisotropic vs. isotropic finale
`batch_galaxy_fit.py`	Multi-galaxy far-field q fits
`batch_sparc_q_fit.py`	138-galaxy fits (NFW comparison)
`knot_model_relative_effect.py`	Knot model simulation vs. observation
`nea_galaxy_fit.py`	Core N.E.A. galaxy fitting
`final_holo_audit_from_table2.py`	Holographic invariant audit ($I = 1.079$)
`nea_extreme_tests.py`	Extreme-scale consistency checks

All code and data available at `https://github.com/hzhangy/nea2gravity`.

Appendix B: Comparison of Three Gravitational Theories

Table 9: Comparison of Newtonian gravity, general relativity, and N.E.A. gravity

Theory	Core equations	Physical meaning
Newtonian gravity	$\mathbf{F} = -G\frac{m_1 m_2}{r^2}\hat{\mathbf{r}}$, $\Phi = -\frac{GM}{r}$	Instantaneous inverse-square force; mass generates potential.
General relativity	$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$, geodesic equation	Gravity as spacetime curvature; matter curves spacetime.
N.E.A. gravity	$V_{\text{obs}}^2 = V_{\text{bar}}^2\left[1 + \left(\frac{R}{R_c}\right)^{2-q}\right]$, $q = 2 - \lambda$, $F \propto 1/r^q$	Gravity as a collective effect of tension; $q \approx 2$ (3D) in high density, $q \approx 1$ (2D) in low density; R_c is the tension radius.

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